A Multi-Objective Optimal Power Flow (OPF) Algorithm for Electric Vehicle Charging to Ensure National Grid Stability in Bangladesh.

Md. Ratul Hasan, Md. Sayed Parvej
Electrical and Electronic Engineering, Jashore University of Science and Technology, Jashore-7408, Bangladesh

Problem Statement

- Context: Bangladesh's grid faces severe instability during evening peak hours (6 PM - 10 PM).
- The Threat: Uncontrolled EV charging coincides with this peak.
- The Consequence: Distribution transformers will overheat and trip, causing localized blackouts.

Objectives

- Grid Stability: Respect transformer thermal limits and line capacity.
- Economic Optimization: Leverage Dynamic Time-of-Use (ToU) pricing (charge when cheap).
- Sustainability: Maximize charging during renewable energy (Solar) availability.

Methodology / System Architecture

- Input Data: Real-time Grid Voltage, EV Battery Status (SoC), Market Price, Solar Irradiance.
- The Controller: The "Multi-Objective OPF Algorithm."
- Output: Regulated Charging Current (I_{ref}) sent to the EV charger.

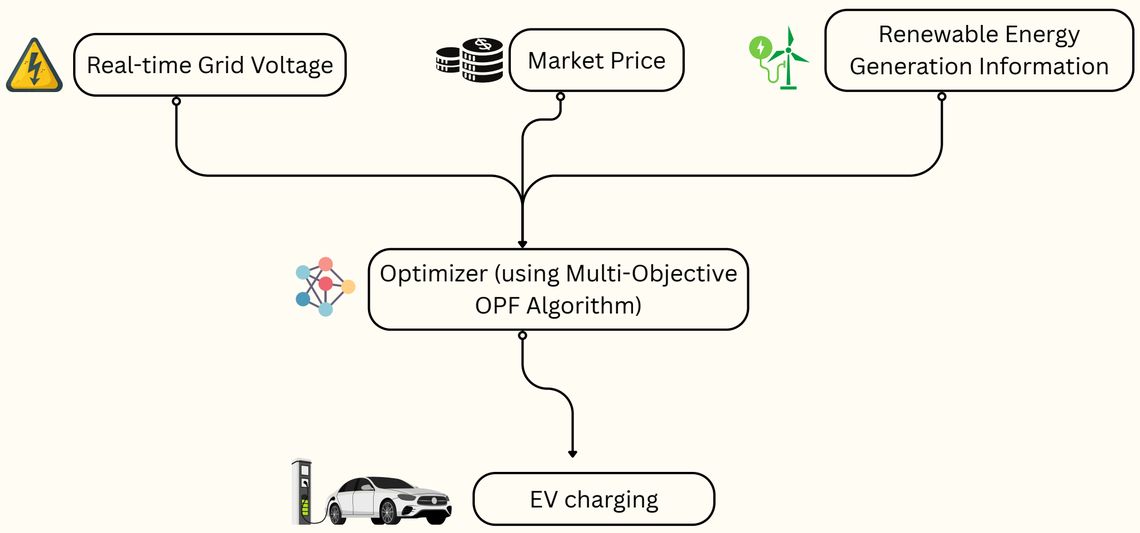


Figure 1: System flow diagram

Mathematical Formulation

$J = \omega_1 (Cost) + \omega_2 (GridStress) - \omega_3 (GreenEnergy)$

Objective: To minimize J

Constraints:

- Voltage Stability: $V_{min} \leq V_i \leq V_{max}$
- Transformer/Line Limit: $S_{flow} \leq S_{max}$
- Power Balance: $P_{gen} = P_{load} + P_{EV} + P_{loss}$
- Charger Limit: $0 \leq P_{EV} \leq P_{rated}$
- User Deadline: $SoC_{departure} \geq SoC_{target}$)

Expected Outcome

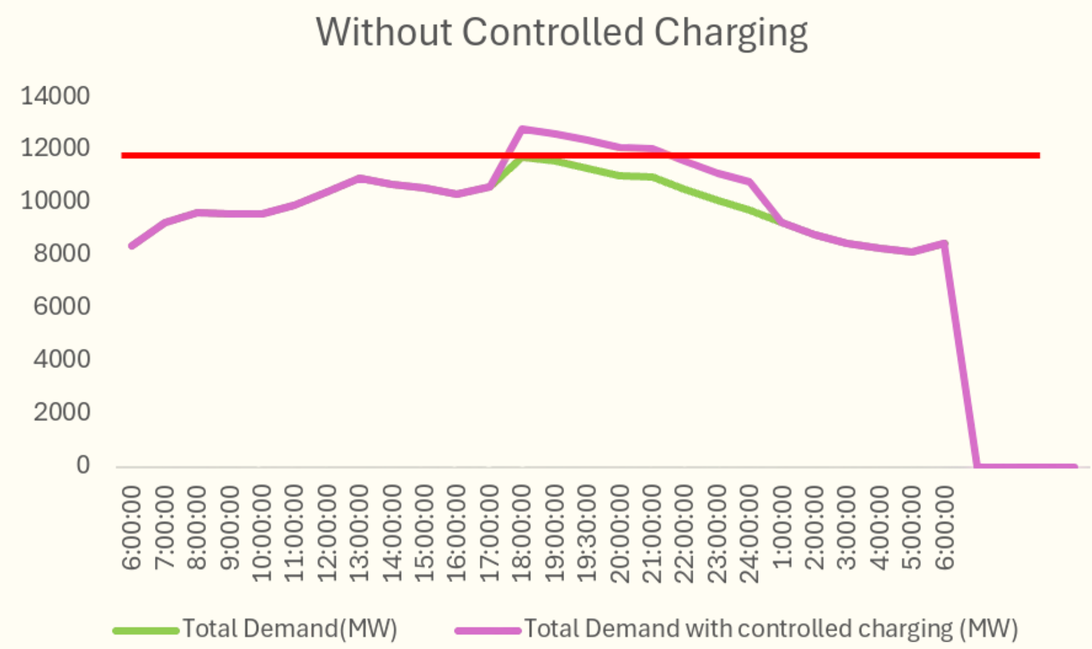


Figure 2: Demand curve for uncontrolled charging

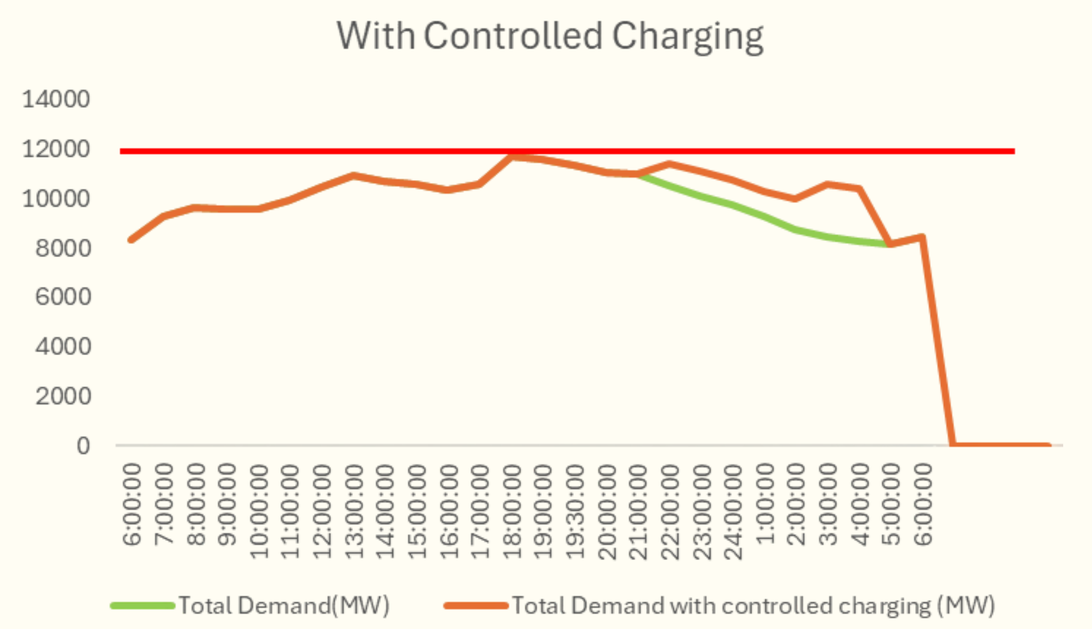


Figure 3: Demand curve for controlled charging

Conclusion & Impact

This work presents a scalable Multi-Objective OPF framework capable of transforming Electric Vehicles from a grid liability into a flexible asset. By intelligently shifting charging loads away from the critical 18:00–21:00 peak window, the system ensures the sustainable integration of the government's targeted 30% EV fleet by 2030.

Implementation Strategy

Implementation Strategy
Leveraging existing smart meters, the system controls IoT-enabled chargers to dynamically prevent overloads. Users set charging deadlines via a mobile app, benefiting from lower off-peak electricity rates. This scalable software solution ensures grid stability without requiring expensive hardware upgrades.

References

1. https://erp.pgcb.gov.bd/w/generations/view_generations
2. https://www.thedailystar.net/business/news/30-govt-vehicles-go-electric-2030-4008391

Dept. of Electrical and Electronic Engineering
Jashore University of Science & Technology,
Jashore-7408, Bangladesh